

AgriVision: Automated Multi-Class Fruit Classification and Quality Assessment Using Transfer Learning

1. Introduction

Agriculture plays a vital role in global food supply chains, and accurate fruit classification is essential for efficient sorting and quality control. Manual inspection is time-consuming and prone to human error. This project presents an automated fruit classification system using computer vision and deep learning techniques. Transfer learning with MobileNetV2 is used to achieve high accuracy while keeping computational costs low.

2. Objectives

The objectives of this project are to understand computer vision fundamentals, analyze a fruit image dataset, apply preprocessing and data augmentation techniques, implement transfer learning using MobileNetV2, evaluate model performance, and develop a practical inference function.

3. Dataset Description

The dataset was sourced from Roboflow and contains labeled images of multiple fruit categories. It is divided into training, validation, and test sets. Images are organized in class-specific folders, allowing automatic label assignment during training.

4. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand class distribution and visual variations in the dataset. Sample images from each class were visualized, and class-wise image counts were analyzed to identify imbalance.

5. Data Preprocessing and Augmentation

All images were resized to 224x224 pixels to match the input requirements of MobileNetV2. Preprocessing consistent with the pretrained model was applied. Data augmentation techniques such as rotation, flipping, zooming, and translation were used on the training set to improve generalization.

6. Model Architecture and Transfer Learning

MobileNetV2, pretrained on ImageNet, was used as the base model. The top classification layers were removed, and a custom classification head was added. The base model was initially frozen to retain learned features.

7. Model Training

The model was trained using the Adam optimizer and categorical cross-entropy loss. Early stopping was employed to prevent overfitting by monitoring validation loss.

8. Model Evaluation

The trained model was evaluated on the test dataset using accuracy, classification reports, and a confusion matrix. The results showed strong generalization with most classes achieving high precision and recall.

9. Error Analysis

Misclassified images were analyzed to understand model limitations. Errors were mainly due to visual similarity between certain fruits, lighting variations, and partial occlusion.

10. Fine-Tuning

Selective fine-tuning of the upper layers of MobileNetV2 with a lower learning rate resulted in a slight improvement in performance.

11. Inference and Deployment

A reusable prediction function was implemented to classify new fruit images and return confidence scores, demonstrating real-world usability.

12. Conclusion

This project successfully demonstrates an end-to-end computer vision pipeline for fruit classification. Transfer learning enabled efficient training and strong performance. Future work may include quality grading and deployment as a web application.

13. Tools and Technologies

Python, TensorFlow, Keras, MobileNetV2, Google Colab, Matplotlib, NumPy, and a Roboflow dataset were used in this project.

14. Acknowledgment

I would like to thank Ms. Syeda Faiza Nasim for her guidance and support throughout the completion of this project.